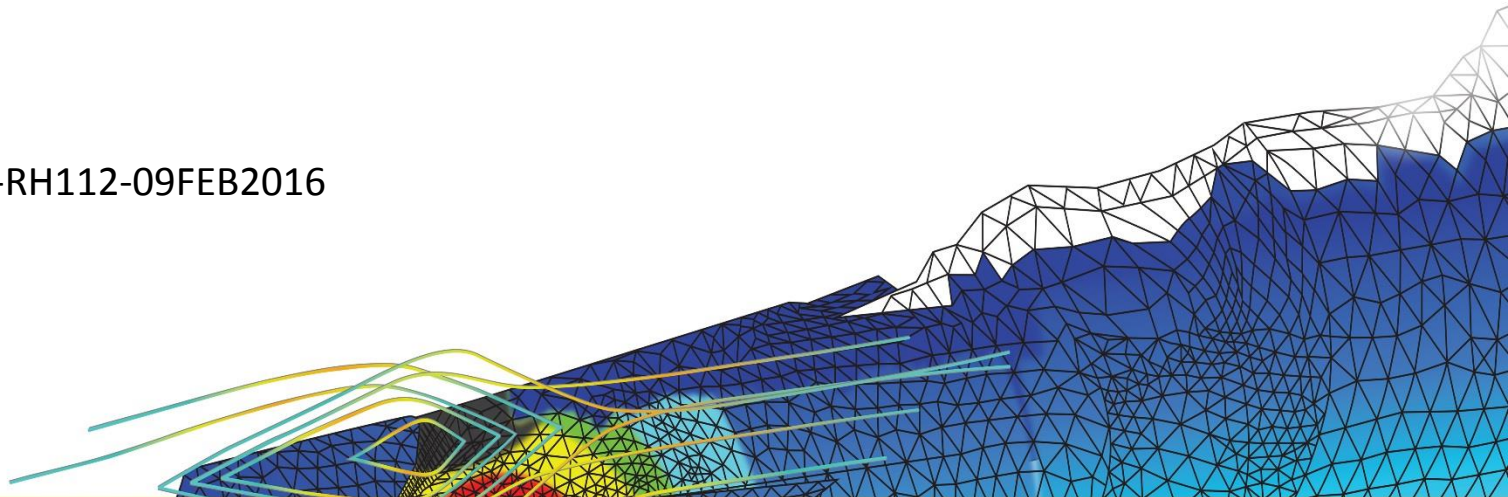




RedHawk Signal EM Analysis

VERSION: V5.1-AL-RH112-09FEB2016



Agenda

- **Key Features**
- **Methodology for Signal EM**
- **RedHawk Signal EM Analysis Flow**
- **Advanced EM rules and settings**
- **Analyzing Signal EM Results**
- **Signal EM Explorer**

Key Features

Usability

- Ease of Use
- Data preparation, GUI interface very similar to RedHawk IR/DvD
- Unified run for avg, rms and peak EM checks. Combined reporting
- Full-chip capacity, Fast Run-times

Accuracy

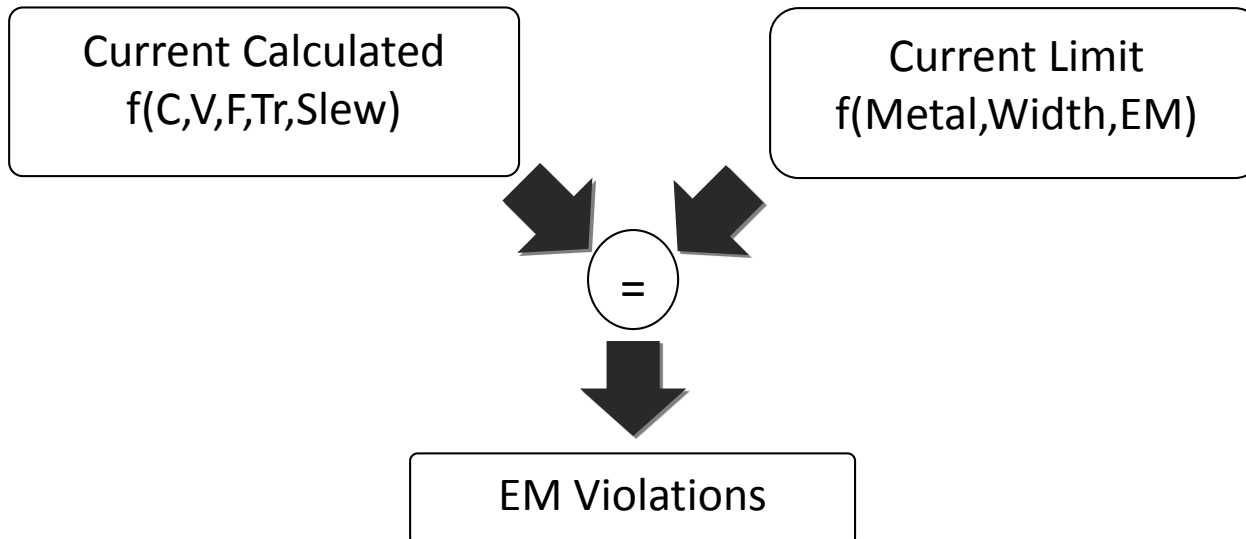
- Accurate estimation of currents from STA and SPEF loading
- Current modeling using polynomial profiles
- Support for all advanced 40/28 nm EM rules

Debug

- GUI based cross probing capability
- RedHawk Explorer based debug
- Selective net analysis and debug
- What-if capability for fixing and analyzing

Methodology for SigEM

- The avg, peak and RMS currents are estimated at each wire segment
- The current limit for each segment is calculated from tech file and wire properties
- $\text{EM violation \%} = \text{Current Calculated} / \text{Current Limit}$



Methodology for SigEM

- The true average, RMS and peak currents are given by

$$I_{avg} = \frac{\int_0^{\tau} |I(t)| dt}{\tau} \quad I_{rms} = \sqrt{\frac{\int_0^{\tau} I^2(t) dt}{\tau}} \quad I_{peak} = \max(|I(t)|)$$

- RedHawk estimates the I_{avg} , I_{rms} and I_{peak} using the Net capacitance, Supply voltage, Toggle rate, Frequency and Slew rate

$$I_{avg} = C \cdot V \cdot f \cdot tr$$

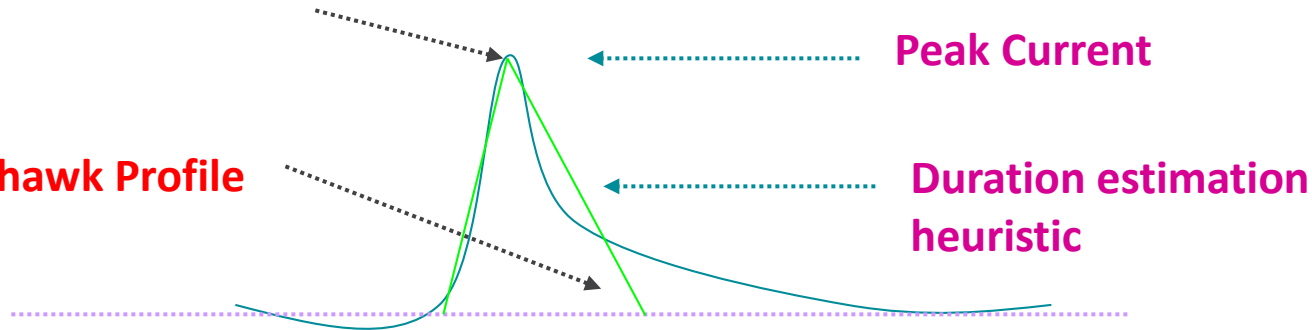
$$I_{rms} = F(C, V, f, tr, slew)$$

$$I_{peak} = F(C, V, slew)$$

Methodology for SigEM: Current Modeling

HSPICE Profile

Redhawk Profile

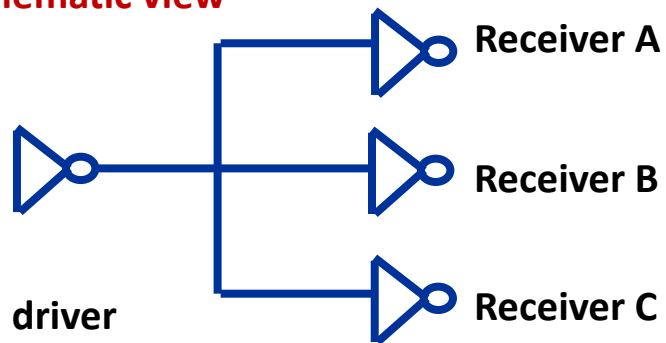


- **Current profile construction:**
 - STA information, signal transitions
 - SPEF rc interconnect
 - R shielding effects for high resistive wires
- **Current calculation:**
 - Avg, RMS currents are based on signal toggle rate
 - Final peak current is based on original profile

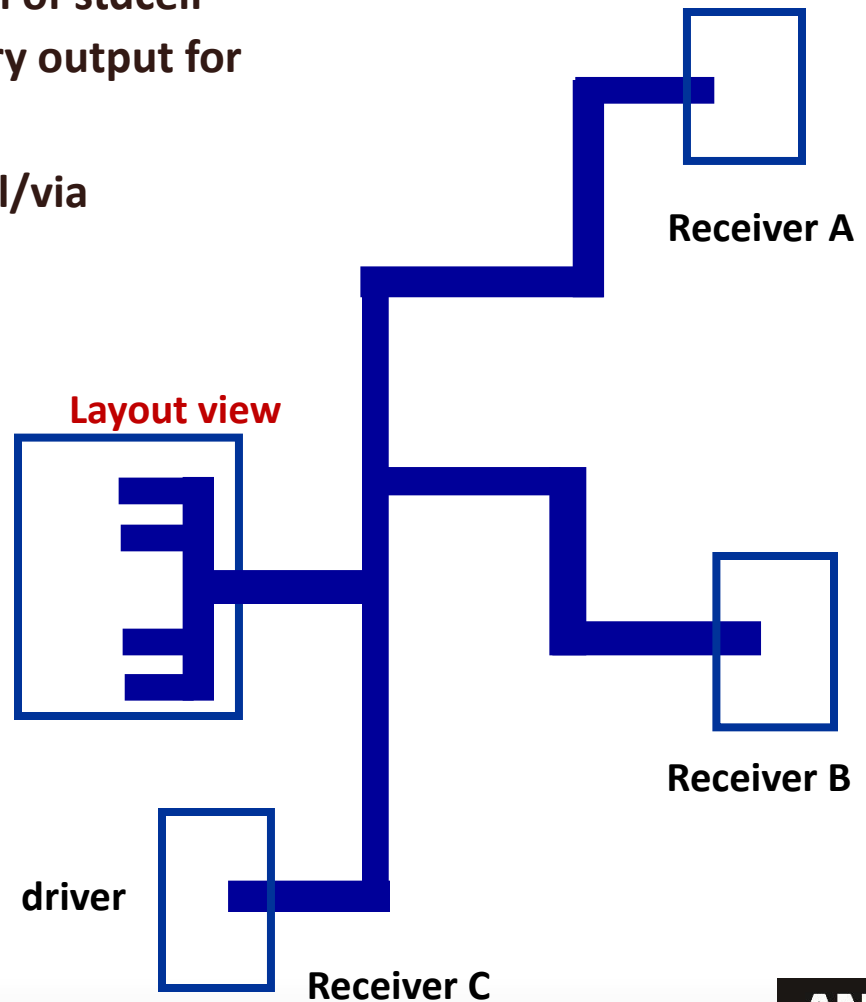
Scope of Inter-Cell Signal EM Analysis

- Analysis is performed from driver pin of stdcell to all receiver pins of stdcells, primary output for each net
- Analysis is performed on every metal/via segment of the nets

Schematic view



Layout view

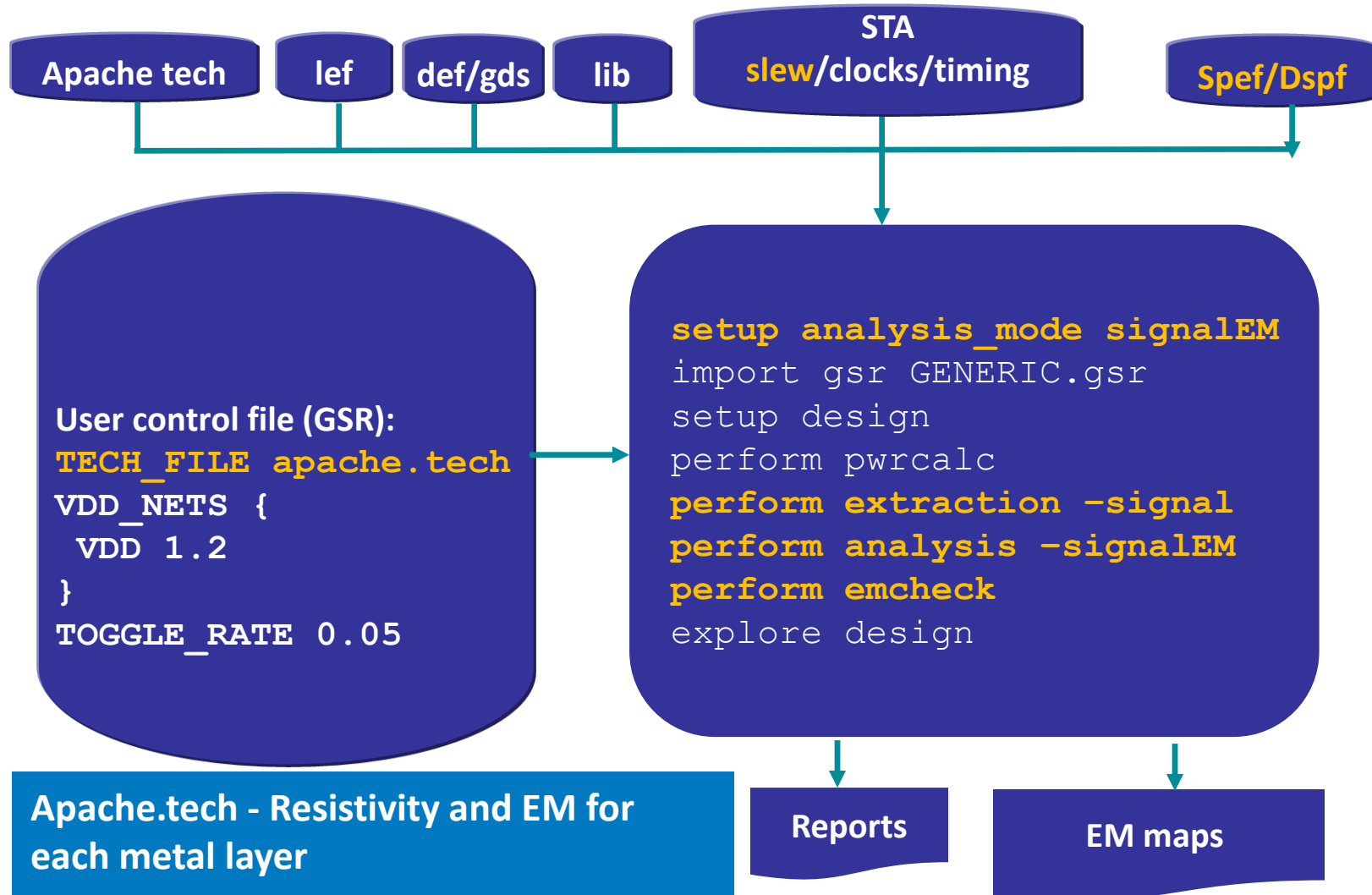


RedHawk Signal EM Analysis Flow

Input Data Requirements

- **Same data preparation as a regular RedHawk Voltage drop Analysis**
 - Routed design netlist in the DEF format
 - .lib for pin capacitance
 - Signal parasitic information in the SPEF/DSPF format
 - Timing information (frequency, slew, clock nets)
 - VCD or instance-specific toggle information [optional]
 - Technology file with EM limits

Signal EM Analysis Flow



Command File Review

Note: The 'setup analysis_mode' step must be provided before 'setup design' for signal EM, as opposed to the power EM analysis procedure, where it is run after 'setup design'.

```
# Import data
setup analysis_mode signalEM
import gsr GENERIC.gsr
setup design

# Calculate power
perform pwrcalc

# extracting signal nets
perform extraction -signal

# Perform Signal EM analysis
perform analysis -signalEM

# creating EM analysis reports
perform emcheck

# result exploration using RHE
explore design
```

Run-time / Memory Summary

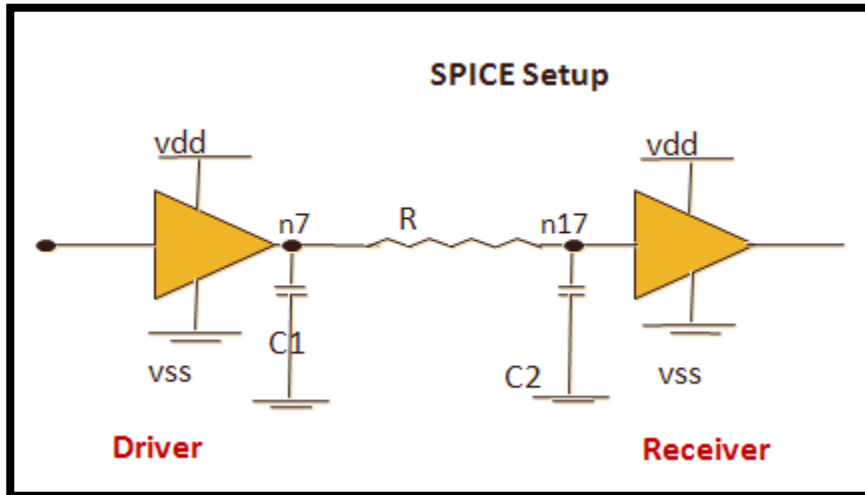
Log Message Viewer <@aesuse01>

Errors/Warnings Summary CPU/Memory Usage Setup Design Power Results

Stages	Memory (MB)	Total CPU Time	Wall Time
Importing LEF	521	00 : 00 : 03	00 : 00 : 06
Importing Library	524	00 : 00 : 04	00 : 00 : 08
Importing DEF	835	00 : 00 : 20	00 : 00 : 25
Setting Up Database	1248	00 : 00 : 57	00 : 01 : 01
Calculating Power	1287	00 : 01 : 13	00 : 01 : 29
Building Connectivity	1394	00 : 01 : 21	00 : 04 : 58
Extracting RC.	1455	00 : 01 : 31	00 : 05 : 09
Analyzing Signal EM	1491	00 : 01 : 39	00 : 05 : 17

Log File:

Recent Spice Correlations '65nm and 40nm Designs'



Current	Max % difference between RH and Spice
lavg	8%
Irms	16%

- 30 nets correlated for 40nm and 65nm designs
- All nets compared for lavg , Irms currents

Guard-banding

- Add a 'safety margin' to your signal EM checks

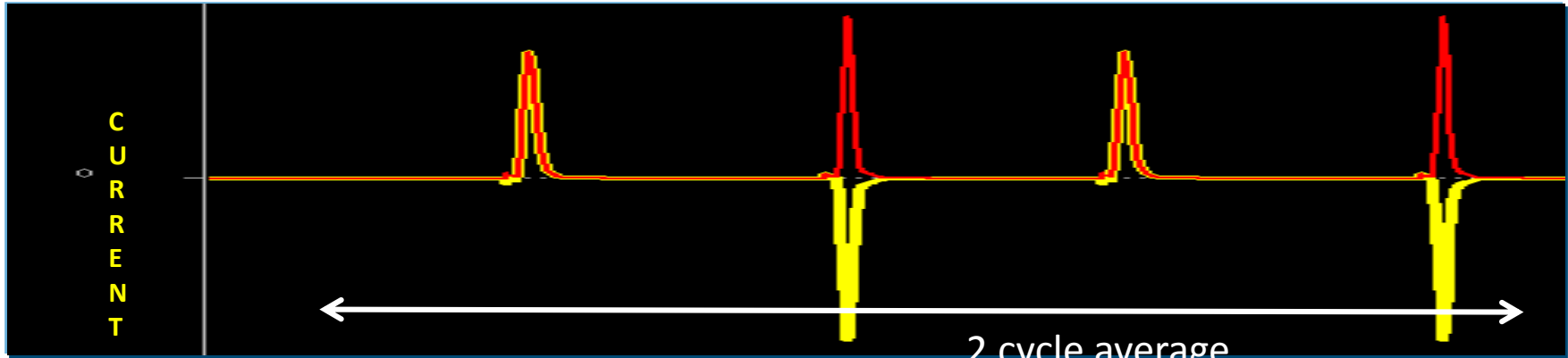
- Example:

Provide 10% safety margin to RMS check, use an EM_SCALE_RMS definition of 1.1

RedHawk will scale up the calculated RMS current by 1.1 and then check against the EM limit.

- **EM_SCALE_AVG 1.03**
(To set a 3% guard band)
- **EM_SCALE_RMS 1.1**
(To set a 10% guard band)
- **EM_SCALE_PEAK 1.05**
(To set a 5% guard band)

Avg Current Methodology

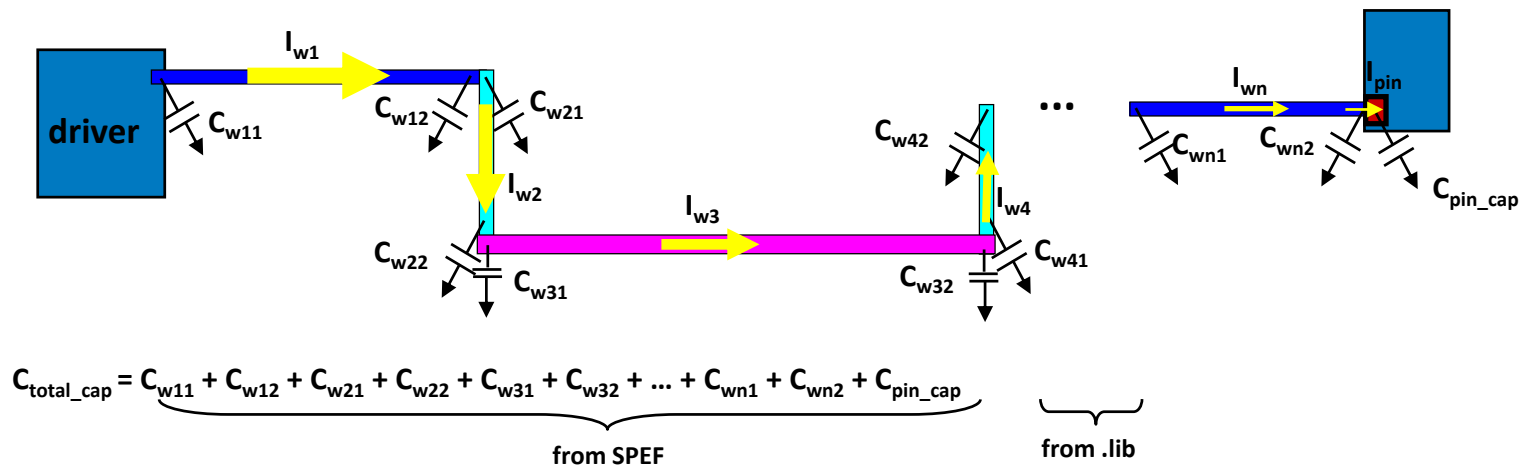


Current at Driver Output

Absolute Value of Current at Driver Output

- For a signal net , since it's a bi-directional current the resultant average is very close to zero
- However Redhawk by default **reports the ABSOLUTE (rectified)** value of the current
 - $I_{avg} = C \cdot V \cdot f \cdot t_r$
- “SEM_RECOVERY_FACTOR <R>” keyword can be used to account for reverse current recovery
 - $I_{avg} = C \cdot V \cdot f \cdot t_r (1-R)/2$

Avg Current Correlation



Due to charging of interconnect capacitors, $I_{w1} > I_{w2} > I_{w3} > I_{w4} \dots > I_{wn} > I_{pin}$

$$I_{avg} \text{ (driver output)} = C_{tot} * V * Freq * TR$$

Advanced EM Rules and Settings

Signal EM Tech file

- Can have single EM rule per metal/via layers
- Length/width dependent EM rule for both metal as well as vias (advanced process nodes). Example below:

```
metal MET2 {
...
  POLYNOMIAL BASED EM_DC {
    LENGTH_RANGES { 5 16 }
    WIDTH_RANGES { 0.15 }
    EM_POLYNOMIALS {
      { (0.04948 + 0.00173/W + 0.92325*W) * (2/L) IMAX 0.137 + 0.00479/W + 2.5593 * W }
      { (0.04948 + 0.00173/W + 0.92325*W) * (2/L) IMAX 0.137 + 0.00479/W + 2.5593 * W }
      { (0.04948 + 0.00173/W + 0.92325*W) * (200/L) IMAX 3 }
      { (0.04948 + 0.00173/W + 0.92325*W) * (200/L) IMAX 3 }
      { (0.04948 + 0.00173/W + 0.92325*W) * (200/L) }
      { (0.04948 + 0.00173/W + 0.92325*W) * (200/L) }
    }
  }
  EM_TEMP_RATING {
    25 1
    100 1.2
  }
  Above ild1_2
}

via VIA1 {
Resistance 18.2000
TNOM_EM 100
EM_TEMP_RATING {
  25 1
  100 0.2
  125 0.1
}
UpperLayer MET2
LowerLayer MET1
  POLYNOMIAL BASED EM_DC {
    COND_RULE { ( Wv == 0.1 AND Lv == 0.1 ) AND ( Lb <= 10 ) AND ( Lu >= 10 ) AND (UPSTREAM)}
    EM_POLYNOMIAL { 4 * 0.1 }
    COND_RULE { ( Wv == 0.1 AND Lv == 0.1 ) AND ( Lb <= 10 ) AND ( Lu >= 10 ) AND (DOWNSTREAM)}
    EM_POLYNOMIAL { 2 * 0.1 }
    COND_RULE { ( Wv == 0.1 AND Lv == 0.1 ) AND ( Lb <= 1000 ) AND ( Lu <= 1000 ) }
    EM_POLYNOMIAL { 1 * 0.1 }
  }
}
```

Polynomial Example on Metal layer

■ *Apache EM Polynomial before v11.2*

```
POLYNOMIAL_BASED_EM_DC{
  LENGTH_RANGES { 10 }
  WIDTH_RANGES { 0.5 }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }

  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }

  EM_POLYNOMIAL { 1 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 2 * 0.2 * ( w - 0.1 ) }
  EM_POLYNOMIAL { 2 * 0.2 * ( w - 0.1 ) }
}

POLYNOMIAL_BASED_EM_PEAK{
  LENGTH_RANGES { 3.402823e+38 }
  WIDTH_RANGES { 3.402823e+38 }
  EM_POLYNOMIAL { 10 * ( w - 0.1 ) }
}

POLYNOMIAL_BASED_EM_RMS{
  LENGTH_RANGES { 3.402823e+38 }
  WIDTH_RANGES { 3.402823e+38 }
  EM_POLYNOMIAL { sqrt ( 1 * delta_T * ( w - 0.1 ) ^ 2 ) }
}
```

■ *Following format is available since v11.2*

```
POLYNOMIAL_BASED_EM_DC{
  COND_RULE { L > 10 AND w < 0.5 }
  EM_POLYNOMIAL { 1 * 0.2 * ( w - 0.1 ) }
  COND_RULE { L > 10 AND w >= 0.5 }
  EM_POLYNOMIAL { 2 * 0.2 * ( w - 0.1 ) }
  COND_RULE { L <= 10 }
  EM_POLYNOMIAL { 3 * 0.2 * ( w - 0.1 ) }
}

POLYNOMIAL_BASED_EM_PEAK{
  COND_RULE { }
  EM_POLYNOMIAL { 10 * ( w - 0.1 ) }
}

POLYNOMIAL_BASED_EM_RMS{
  COND_RULE { }
  EM_POLYNOMIAL { sqrt ( 1 * delta_T * ( w - 0.1 ) ^ 2 ) }
}
```

1. *No need to specify the LENGTH/WIDTH_RANGES in the polynomial EM equation since v11.2*
2. *RH will read content of COND_RULE , then calculate corresponding EM rule*

Polynomial Example on Via layer

- *Following format for via layer is available since v11.2*

```
POLYNOMIAL_BASED_EM_DC {  
  COND_RULE { ( Wv == 0.5 AND Lv == 0.5 ) AND ( Lb <= 10 ) AND ( Lu <= 10 ) }  
  EM_POLYNOMIAL { 3 * 0.1 }  
  COND_RULE { ( Wv == 0.5 AND Lv == 0.5 ) AND ( Lb > 10 AND Wb >= 2 ) AND ( Lu  
    <= 10 ) }  
  EM_POLYNOMIAL { 2 * 0.1 }  
  COND_RULE { ( ( Wv == 0.5 AND Lv == 0.5 ) AND ( NOT ( ( Lb <= 10 ) AND ( Lu  
    <= 10 ) ) AND ( ( Lb > 10 AND Wb >= 2 ) AND ( Lu <= 10 ) ) ) ) }  
  EM_POLYNOMIAL { 1 * 0.1 }  
  COND_RULE { ( Wv == 1 AND Lv == 1 ) AND ( Lb <= 10 ) AND ( Lu > 10 AND Wu  
    >= 10 ) }  
  EM_POLYNOMIAL { 3 * 0.4 }  
  COND_RULE { ( Wv == 1 AND Lv == 1 ) AND ( Lb > 10 AND Wb >= 2 ) AND ( Lu > 10  
    AND Wu >= 2 ) }  
  EM_POLYNOMIAL { 2 * 0.4 }  
}
```

User can give condition with the combination with following parameters

L/W: Metal Length/Width

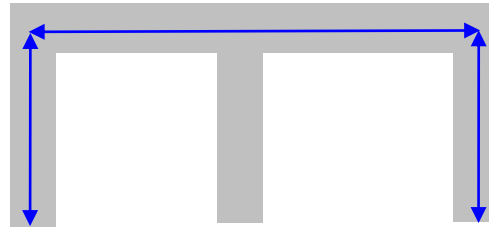
Lv and Wv is Length/Width of via (post-shrink size if it is half node process)

Lu/Wu: Length/Width of upper metal layer connecting to this via

Lb/Wb: Length/Width of below metal layer connecting to this via

Length Calculation for EM Rule Look Up

- GSR keyword "EM_LENGTH_USE_MAX_LENGTH 0|1" to calculate the length for EM rule lookup
 - When "0", RH will use total metal length for EM rule look up
 - When "1", RH will use the longest metal length for EM rule look up.
- The longest length of metal will be calculated like shown below:



Signal EM tech File

- Instead of single tech file containing EM rules for all modes (avg, rms, peak), user can also specify mode specific tech files:
 - **EM_Tech_RMS** <tech_file_nameA>
 - **EM_Tech_AVG** <tech_file_nameB>
 - **EM_Tech_PEAK** <tech_file_nameC>
- The additional tech files should only contain EM limits. For eg:
 - metal xxx { EM number; polynomial_based_* ; width_based_em; etc }
 - via xxx { EM number; polynomial_based_* ; emv_vs_area; etc }

Drawn vs Silicon Width Based Analysis

- Controlled by following GSR keywords:

USE_DRAWN_WIDTH_FOR_EM

By default RedHawk uses the drawn width to calculate current density for wires. When this option is turned off, silicon width is used for current density calculation. *DMP compatible. Optional. Default: the default setting for this keyword is foundry-dependent. Please consult Apache Support for appropriate usage.*

Syntax:

USE_DRAWN_WIDTH_FOR_EM [0 | 1]

	USE_DRAWN_WIDTH_FOR_EM_LOOKUP 1	USE_DRAWN_WIDTH_FOR_EM_LOOKUP 0
USE_DRAWN_WIDTH_FOR_EM 1	W (lookup) = W(D) W=W(D)	Not Supported
USE_DRAWN_WIDTH_FOR_EM 0	W (lookup) = W(D) W=W(S)	W (lookup)=W(S) W=W(S)

Some foundries require EM to be computed based on silicon width, but lookup based on drawn width for 20 nm and lower nodes. This flow is enabled by setting the two keywords as shown in the table above.

USE_DRAWN_WIDTH_FOR_EM_LOOKUP

By default, uses drawn wire width for the EM rule lookup table, rather than silicon width (when set to 0). *DMP compatible. Optional. Default: the default setting for this keyword is foundry-dependent. Please consult Apache Support for appropriate usage.*

Syntax:

USE_DRAWN_WIDTH_FOR_EM_LOOKUP [0 | 1]

Example1: Using Drawn Width in analysis

$W(d)$ = Drawn width

- Drawn width is equal to the layout width of the net.
- Hence all nets (of same metal layer) with same width will have same EM limits.

Current Limit for a Net = $W(d) * \text{EM limit from the tech file}$

adsRpt/SignalEM/*.rpt

#layer	#end-to-end_coordinates	#EM_Ratio	#net	#width	#current	#current_limit	#direction
--------	-------------------------	-----------	------	--------	----------	----------------	------------

M3	(645,945,151,650 646,075,151,650)	99.9991%	lpcie_lgc_con/cnm/tlm2cnm_aer_fifo_dout[136]	0.050	1.29149e-05	1.2915e-05	Bidir
M3	(848,875,185,150 849,810,185,150)	99.999%	lpcie_lgc_dbg_oq_head[2]	0.050	1.29149e-05	1.2915e-05	Bidir
M4	(381,315,211,990 381,395,211,990)	99.9998%	n1022_L7_N39	0.050	1.27348e-05	1.2735e-05	Bidir
M4	(313,950,306,325 313,950,308,775)	99.9998%	user_clk2_N_cbuf__L3_N2	0.050	1.27348e-05	1.2735e-05	Bidir
M3	(741,260,374,225 741,260,374,475)	99.9994%	attr_vsec_cap_nextptr[1]	0.050	1.29148e-05	1.2915e-05	Bidir
M5	(488,385,226,350 509,885,226,350)	99.9998%	lpcie_lgc_plm/v51_ngt/FE_OFN5549_ir_pipe_rx6_data_7_	0.050	1.26447e-05	1.2645e-05	Bidir
M4	(260,390,323,525 260,390,324,875)	99.9979%	n1022_L7_N7	0.050	1.27347e-05	1.2735e-05	Bidir

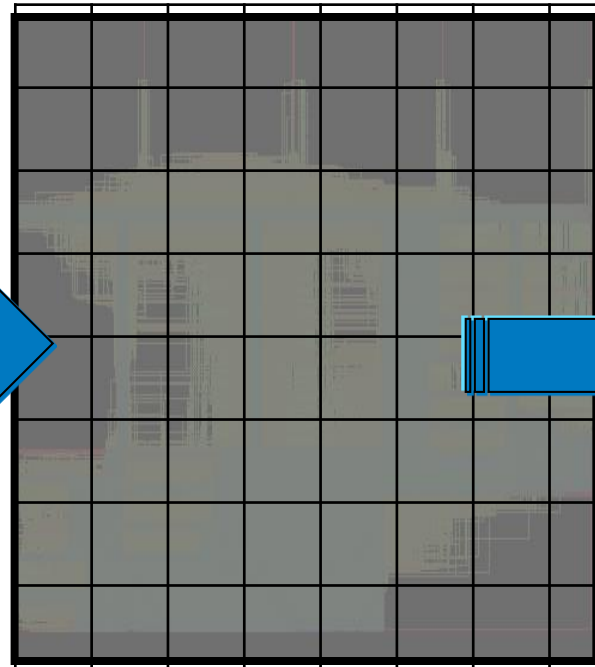
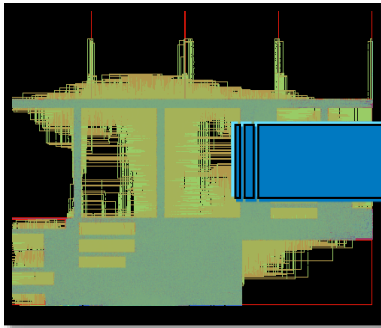
With drawn width based analysis , same EM limits for M3 with similar width

Example2: Using Silicon Width in analysis (Default)

$$W(s) = W(d) - EM_adjust - \text{Etch Factor}$$

Grid creation 10ux10u

Layout



Redhawk Calculates
Width and Spacing Density
for each region

NOTE: Since etching is a function of both metal density and space density in a region, its very difficult to calculate the limit manually.

Parameters Affecting Silicon Width

- **ETCH_VS_WIDTH_AND_SPACING**
- **ETCH_FROM_TOP**
- **HALF_NODE_SCALE_FACTOR**
- **LEF/DEF SCALE FACTOR**
- **EM_ADJUST**

GSR Settings when using TSMC EM Rules

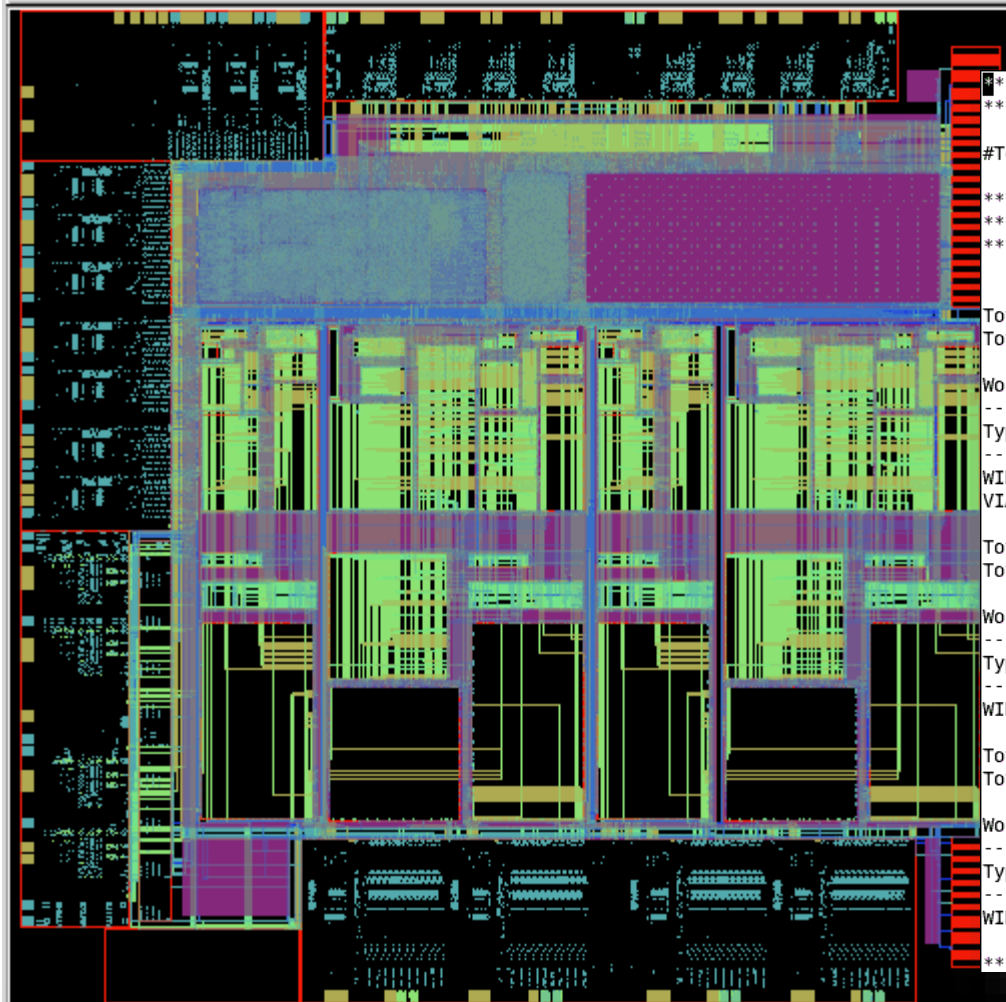
GSR parameter	DC mode	RMS mode	Peak mode	Purpose
TEMPERATURE_EM	✓	✓	✓	Set operating temperature of EM DC mode.
USE_DRAWN_WIDTH_FOR_EM_LOOKUP	✓	✓	✓	Use drawn width when doing EM limitation calculation from tech file (must have if use TSMC EM rule)
USE_DRAWN_WIDTH_FOR_EM	✓	✓	✓	Use drawn width to do EM analysis (must have if TSMC EM rule)
ENABLE_BLECH	✓	✓	✓	Handle length based polynomial equation of tech file (must have if use tech file from iRCX)
DELTA_T_RMS_EM	✗	✓	✗	Set ΔT for RMS EM check
EM_LENGTH_USE_MAX_LENGTH	✓ (TSMC Process should use "1")			"1": Longest path of metal wire "0": total length of metal wire, (Default mode). Available since v11.2

ANALYZING SIGNAL EM RESULTS

Analyzing Signal EM Results

Layout view with all signal wires

adsRpt/redhawk.log



```
*** Finish Analyzing Signal EM
*****
#Tcl: perform emcheck
*****
**** Start EM check
*****

Total number of metal EM violations: 226
Total number of via EM violations: 105

Worst AVG EM violation:
-----
Type  value net  EM_limit  location  name
-----
WIRE  1393.08% net_17882  0.22mA (2769.903,4076.015) METAL1
VIA   533.09% net_17882  0.57mA (2771.730,4042.395) via1

Total number of metal EM violations: 1000
Total number of via EM violations: 0

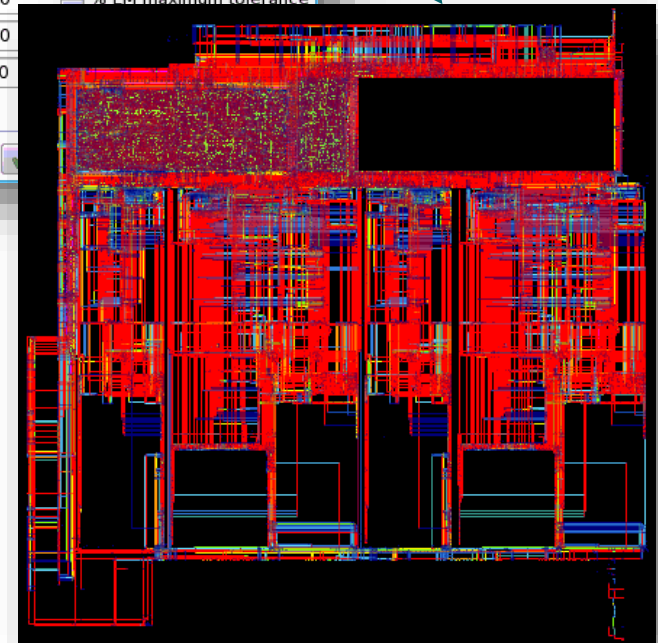
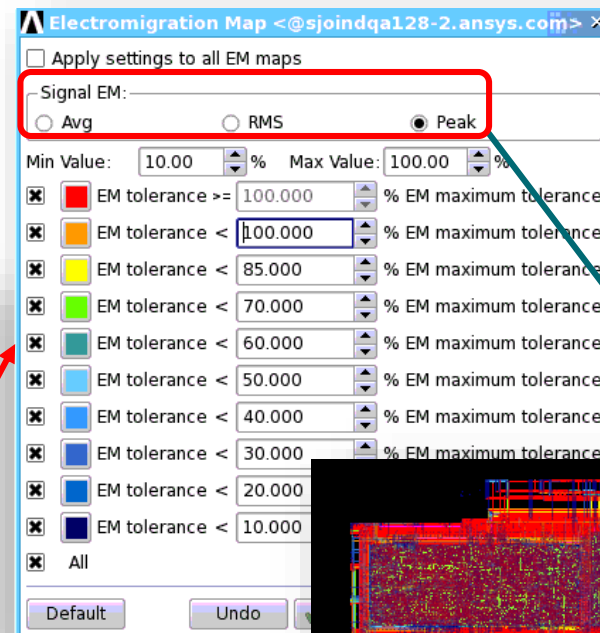
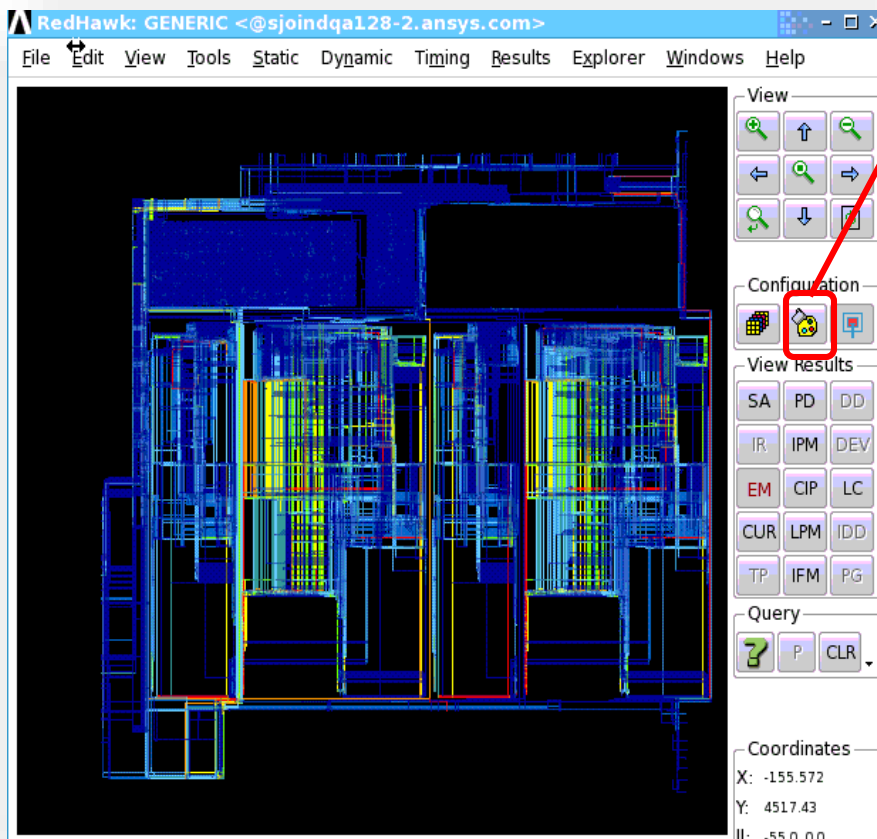
Worst PEAK EM violation:
-----
Type  value net  EM_limit  location  name
-----
WIRE  6132240.00% inst_129228/net_74352  0.35mA (1102.160,3751.705) METAL2

Total number of metal EM violations: 1000
Total number of via EM violations: 0

Worst RMS EM violation:
-----
Type  value net  EM_limit  location  name
-----
WIRE  6993.45% inst_129228/net_74352  0.35mA (1102.160,3751.705) METAL2
*****
```

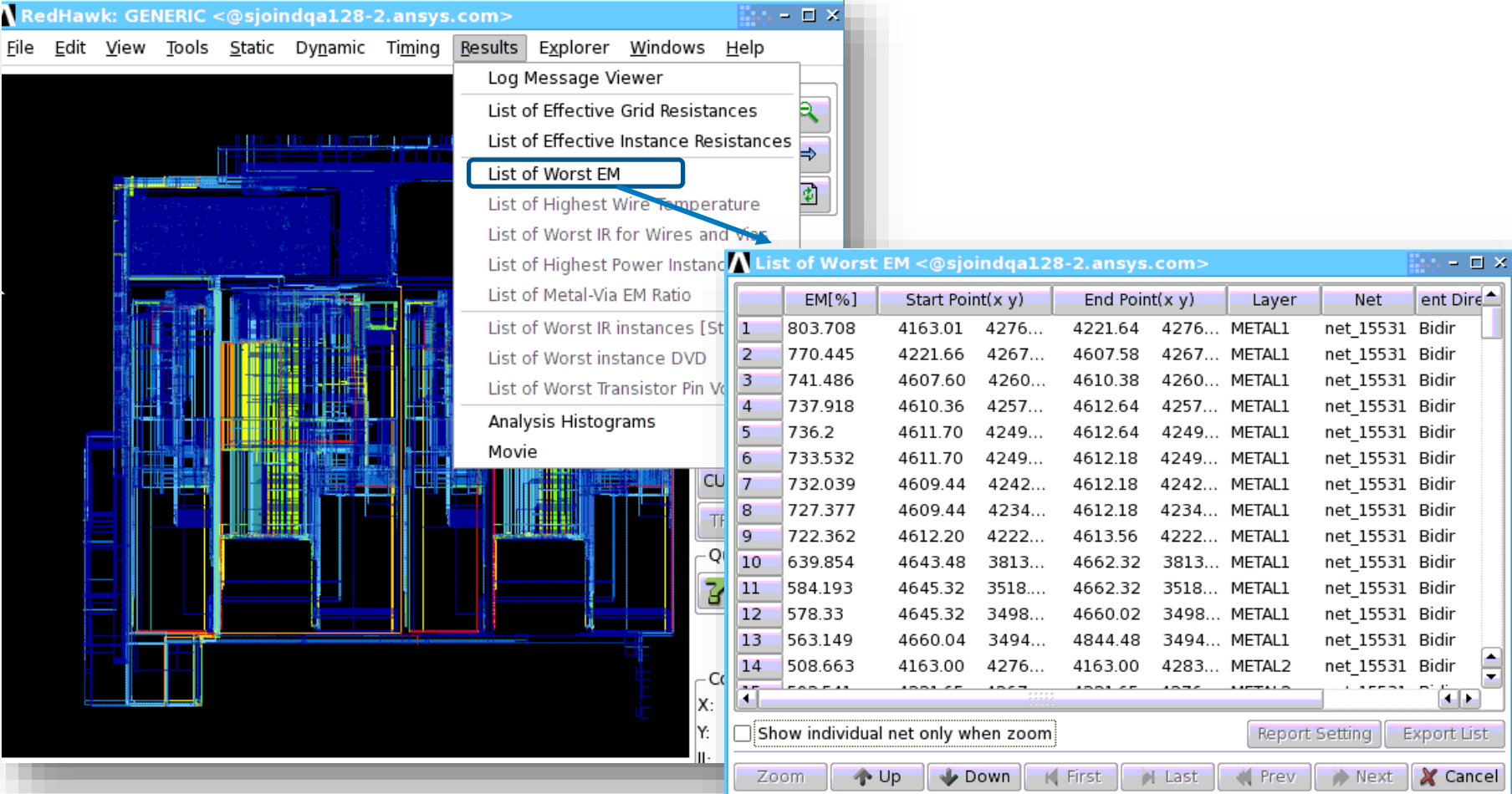
Analyzing Signal EM Results (cont'd)

Click the EM Color map and switch between different modes (RMS, Peak, Average) using radio button (in set color range) to get corresponding EM map



Analyzing Signal EM Results (cont'd)

List of Worst EM report



RedHawk: GENERIC <@sjoindqa128-2.ansys.com>

File Edit View Tools Static Dynamic Timing Results Explorer Windows Help

Log Message Viewer
List of Effective Grid Resistances
List of Effective Instance Resistances
List of Worst EM
List of Highest Wire Temperature
List of Worst IR for Wires and Vias
List of Highest Power Instance
List of Metal-Via EM Ratio
List of Worst IR instances [St
List of Worst instance DVD
List of Worst Transistor Pin V
Analysis Histograms
Movie

List of Worst EM <@sjoindqa128-2.ansys.com>

	EM[%]	Start Point(x y)	End Point(x y)	Layer	Net	ent Dire
1	803.708	4163.01 4276...	4221.64 4276...	METAL1	net_15531	Bidir
2	770.445	4221.66 4267...	4607.58 4267...	METAL1	net_15531	Bidir
3	741.486	4607.60 4260...	4610.38 4260...	METAL1	net_15531	Bidir
4	737.918	4610.36 4257...	4612.64 4257...	METAL1	net_15531	Bidir
5	736.2	4611.70 4249...	4612.64 4249...	METAL1	net_15531	Bidir
6	733.532	4611.70 4249...	4612.18 4249...	METAL1	net_15531	Bidir
7	732.039	4609.44 4242...	4612.18 4242...	METAL1	net_15531	Bidir
8	727.377	4609.44 4234...	4612.18 4234...	METAL1	net_15531	Bidir
9	722.362	4612.20 4222...	4613.56 4222...	METAL1	net_15531	Bidir
10	639.854	4643.48 3813...	4662.32 3813...	METAL1	net_15531	Bidir
11	584.193	4645.32 3518...	4662.32 3518...	METAL1	net_15531	Bidir
12	578.33	4645.32 3498...	4660.02 3498...	METAL1	net_15531	Bidir
13	563.149	4660.04 3494...	4844.48 3494...	METAL1	net_15531	Bidir
14	508.663	4163.00 4276...	4163.00 4283...	METAL2	net_15531	Bidir
15	508.541	4221.66 4267...	4221.66 4276...	METAL1	net_15531	Bidir

☐ Show individual net only when zoom

Report Setting Export List

Zoom Up Down First Last Prev Next Cancel

Analyzing Signal EM Results (cont'd)

- Separate report for rms/peak/average EM violations, in adsRpt/SignalEM

```
aesuse01 GENERIC_tutorial/signalEM> ls -lrt adsRpt/SignalEM/
total 1968
-rw-r--r-- 1 aleena ae      2514 2012-02-28 22:19 connectivity_issue.rpt
-rw-r--r-- 1 aleena ae 1741638 2012-02-28 22:19 GENERIC.droppedSignalNets
-rw-r--r-- 1 aleena ae   31528 2012-02-28 22:19 GENERIC.em.worst.avg
-rw-r--r-- 1 aleena ae  112007 2012-02-28 22:20 GENERIC.em.worst.peak
-rw-r--r-- 1 aleena ae 104384 2012-02-28 22:20 GENERIC.em.worst.rms
lrwxrwxrwx 1 aleena ae      20 2012-02-28 22:20 GENERIC.em.worst -> GENERIC.em.worst.rms
```

- adsRpt/SignalEM/GENERIC.em.worst.rms
 - Reports the highest 1000 J_{rms} (RMS current density) violations.

```
# EM MODE is RMS
# This file reports the EM violations, i.e. EM Ratio = Actual Current Density/Current Density Limit > 100% (default or
from "em_report_percentage") for wire pieces and vias in decreasing order. Unit used for coordinates and dimensions is
um.

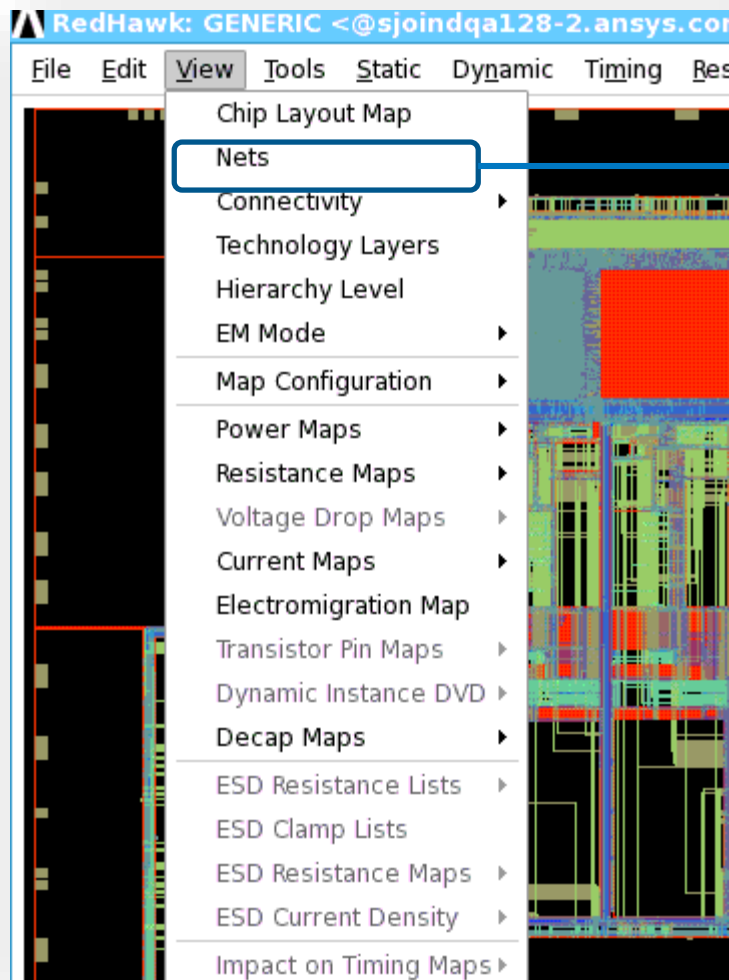
# For wires: #layer #end-to-end_coordinates #EM_Ratio #net #width #current #limit #direction
# For vias: #via_name #x-y_coordinates #EM_Ratio #net #current #limit #direction

METAL2 (1102.160,3751.705 1102.160,3755.358) 6993.45% inst_129228/net_74352 0.200 0.0241531 0.000345368 Bidir
METAL2 (1102.360,3751.705 1102.420,3751.705) 6943.47% inst_129228/net_74352 0.200 0.0239805 0.000345368 Bidir
METAL2 (1102.620,3742.302 1102.620,3751.705) 6823.59% inst_129228/net_74352 0.200 0.0235665 0.000345368 Bidir
METAL3 (1102.648,3742.275 1136.632,3742.275) 6286.69% inst_129228/net_74352 0.200 0.0217122 0.000345368 Bidir
METAL2 (1136.660,3741.045 1136.660,3742.247) 5849.49% inst_129228/net_74352 0.200 0.0202023 0.000345368 Bidir
METAL2 (1136.660,3729.565 1136.660,3741.045) 5372.98% inst_129228/net_74352 0.200 0.0185566 0.000345368 Bidir
```

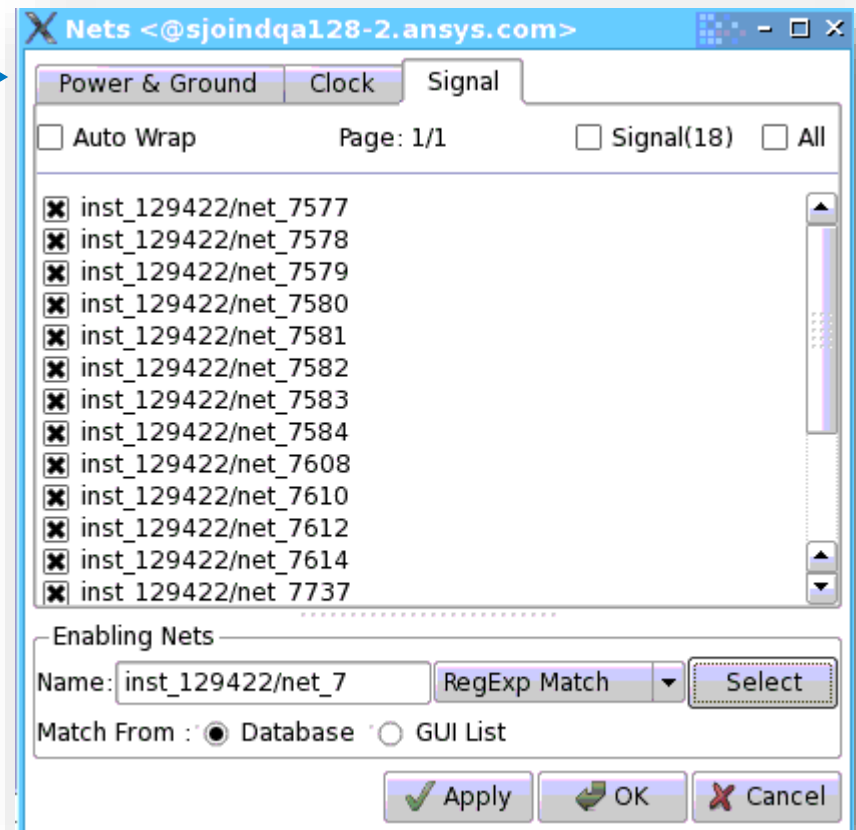
- adsRpt/apache.sigem.info
 - Check signal nets that have no loading capacitance, timing, or driver instance information.

Analyzing Signal EM Results (cont'd)

“EM” display → Select nets to display

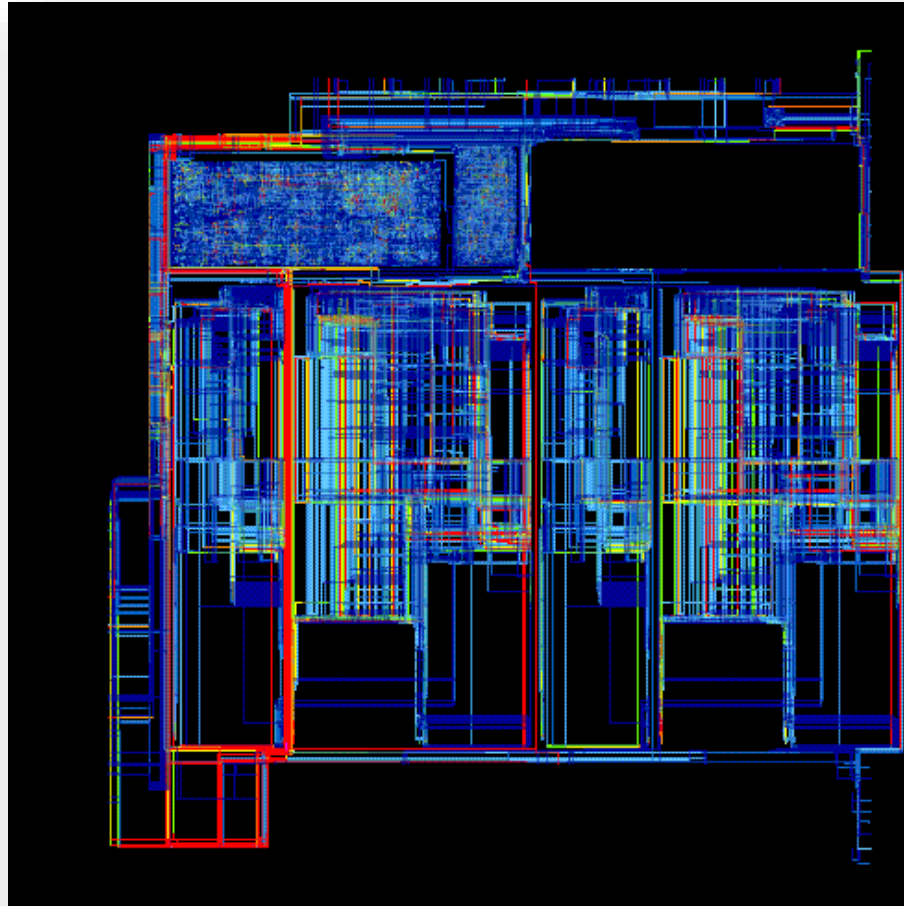


Lists all the nets(clocks and signals in design)



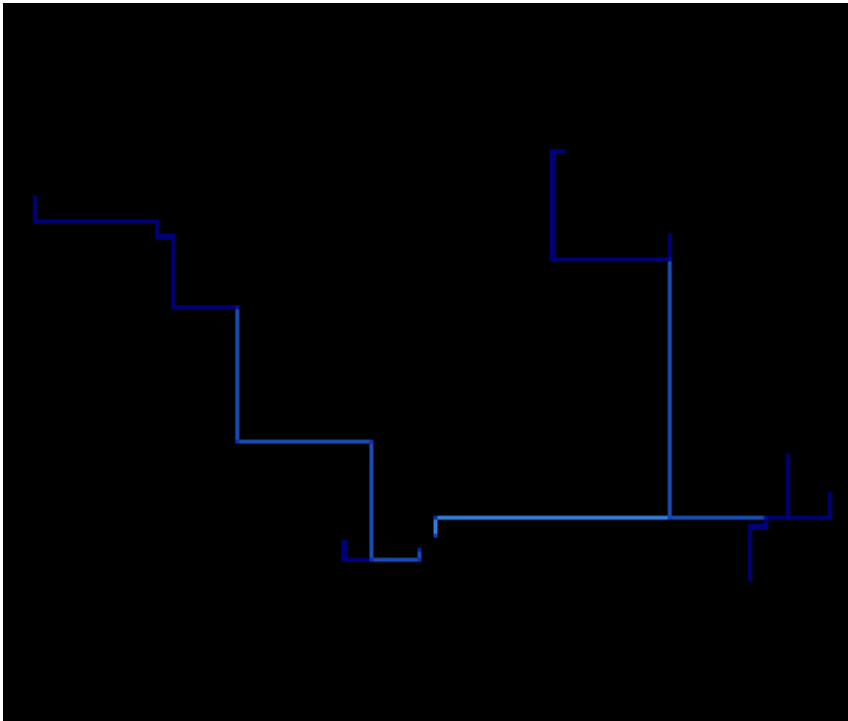
Analyzing Signal EM Results (*cont'd*)

EM Map for Clock Nets (em_mode rms)

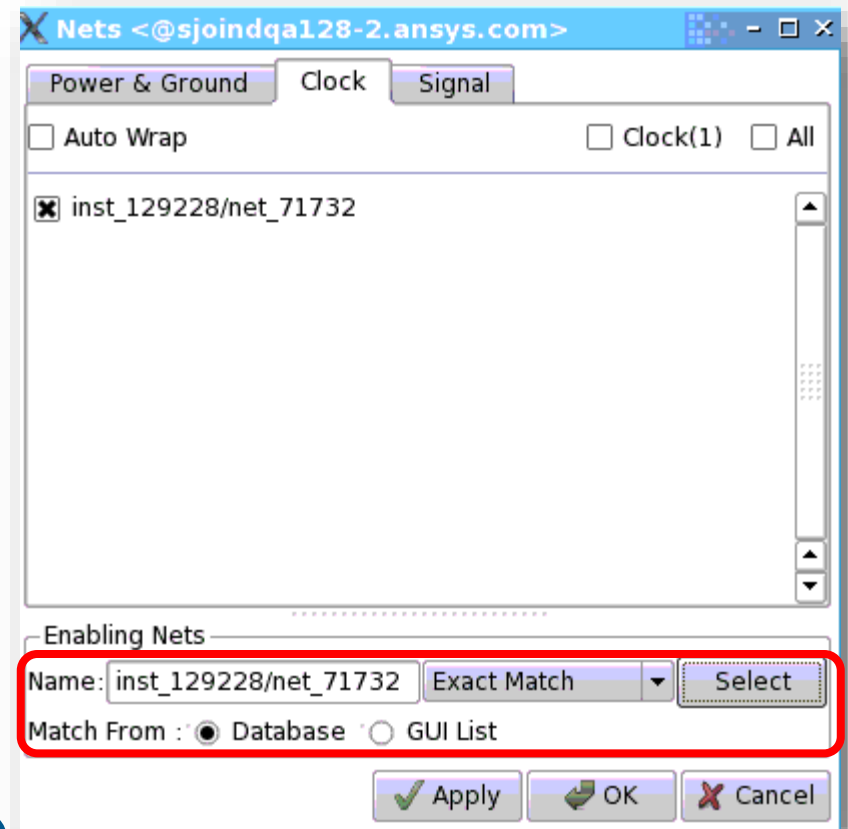


Analyzing Signal EM Results (cont'd)

EM Map for only one Net (inst_129228/net_71732)

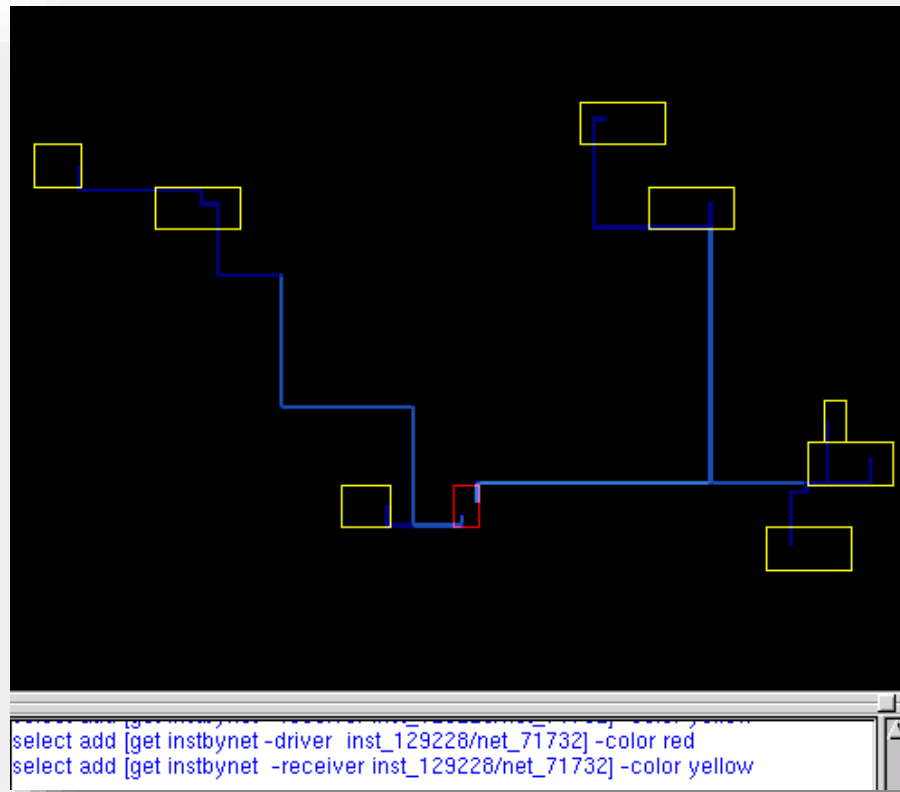


1. Unselect "All" checkbox to unselect all nets
2. Enable Net By Name → inst_129228/net_71732



Analyzing Signal EM Results (*cont'd*)

To highlight Drivers and Receivers for net inst_129228/net_71732

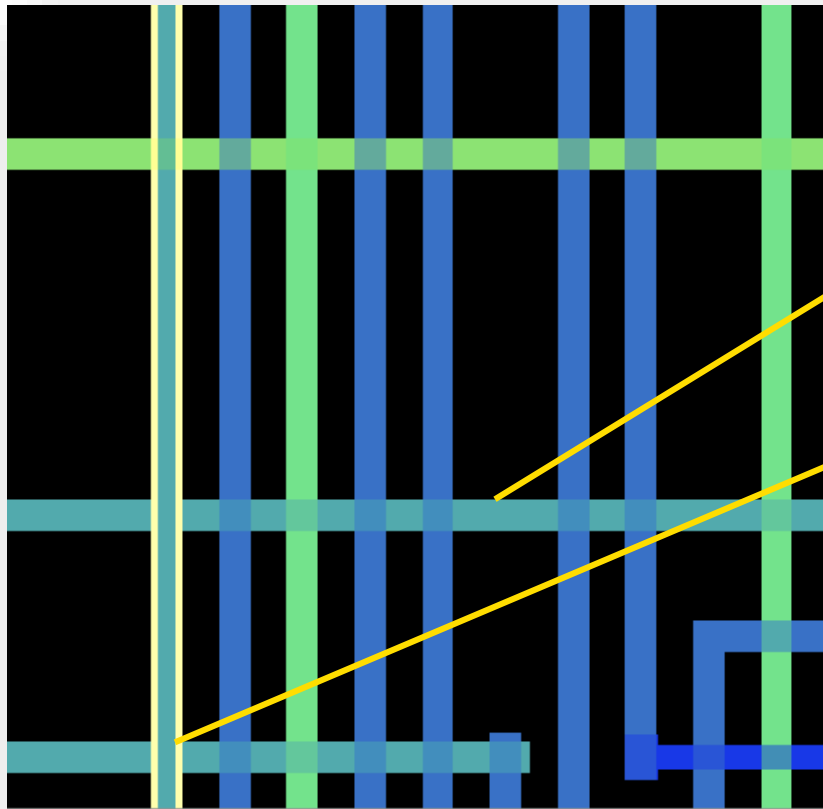


In GUI command window, use “select” and “get instbynet” to highlight :

1. select add [get instbynet -driver inst_129228/net_71732] -color red
2. select add [get instbynet -receiver inst_129228/net_71732] -color yellow

Analyzing Signal EM Results (*cont'd*)

zoom rect 2223 4007 2229 4013



zoom rect 2223.63793945 4007.60888672 2229.24804688 4013.17407227

Net: inst_129228/net_91942
Wire: METAL3 [Id = 232770]
Width: 0.2 um
Length: 8.48 um
Coordinates: (2221.24, 4009.495) (2229.72, 4009.495) (2229.72, 4009.695) (2221.24, 4009.695)
Driver(s) : (inst_129228/inst_465985:Q)
Resistance: 3.89 Ohm
Querypoint (2227.0590, 4009.6220)
Current parallel to wire direction at query position: 2.59426e-05 A <
Current type: Bidir

EM mode: rms
EM rule: const based (EMCurrentDensity: 0.001877 A/um, width: 0.184 um)
EM percentage: 7.51% (EM current limit: 0.000345368 A)

Net: inst_129228/net_93194
Wire: METAL2 [Id = 244540]
Width: 0.2 um
Length: 7.97 um
Coordinates: (2224.46, 4005.805) (2224.66, 4005.805) (2224.66, 4013.775) (2224.46, 4013.775)
Driver(s) : (inst_129228/inst_466908:Y)
Resistance: 3.65 Ohm
Querypoint (2224.5680, 4008.4210)
Current parallel to wire direction at query position: 1.68448e-06 A ^
Current type: Bidir

EM mode: rms
EM rule: const based (EMCurrentDensity: 0.001877 A/um, width: 0.184 um)
EM percentage: 0.49% (EM current limit: 0.000345368 A)

Analyzing Signal EM Results (*cont'd*)

- Current direction and EM violations
- After EM analysis, clicking on any METAL or VIA geometry also shows current direction alongside current

```
Net: inst_129228/net_91942  
Wire: METAL3 [ Id = 232770 ]  
Width: 0.2 um  
Length: 8.48 um  
Coordinates: (2221.24, 4009.495) (2229.72, 4009.495) (2229.72, 4009.695) (2221.24, 4009.695)  
Driver(s) : ( inst_129228/inst_465985:Q )  
Resistance: 3.89 Ohm  
Querypoint (2227.0590, 4009.6220)  
Current parallel to wire direction at query position: 2.59426e-05 A <  
Current type: Bidir  
  
EM mode: rms  
EM rule: const based (EMCurrentDensity: 0.001877 A/um, width: 0.184 um)  
EM percentage: 7.51% (EM current limit: 0.000345368 A)
```

- The symbol conventions followed for displaying the current direction are as shown below:

```
> : For metals, current flow from left to right.  
< : For metals, current flow from right to left.  
^ : For metals, current flow from bottom to top.  
V : For metals, current flow from top to bottom  
Up : For vias, current flow from lower metal layer to upper metal layer  
Down : For vias, current flow from upper metal layer to lower metal layer.
```

Debugging Tips

- **EM_DEBUG 1**
 - Generate .apache/em.debug
 - Format:
net voltage freq toggle duration P1 total_cap pin_cap driver cell
- **EM_DEBUG_NET netname**
 - Only process one net for quick probing and fast turnaround
 - To perform incremental debugging (net by net) in the GUI or a TCL shell, use the following:

setup design

gsr set EM_DEBUG_NET {net1}

perform analysis –signalEM

gsr set EM_DEBUG_NET {net2}

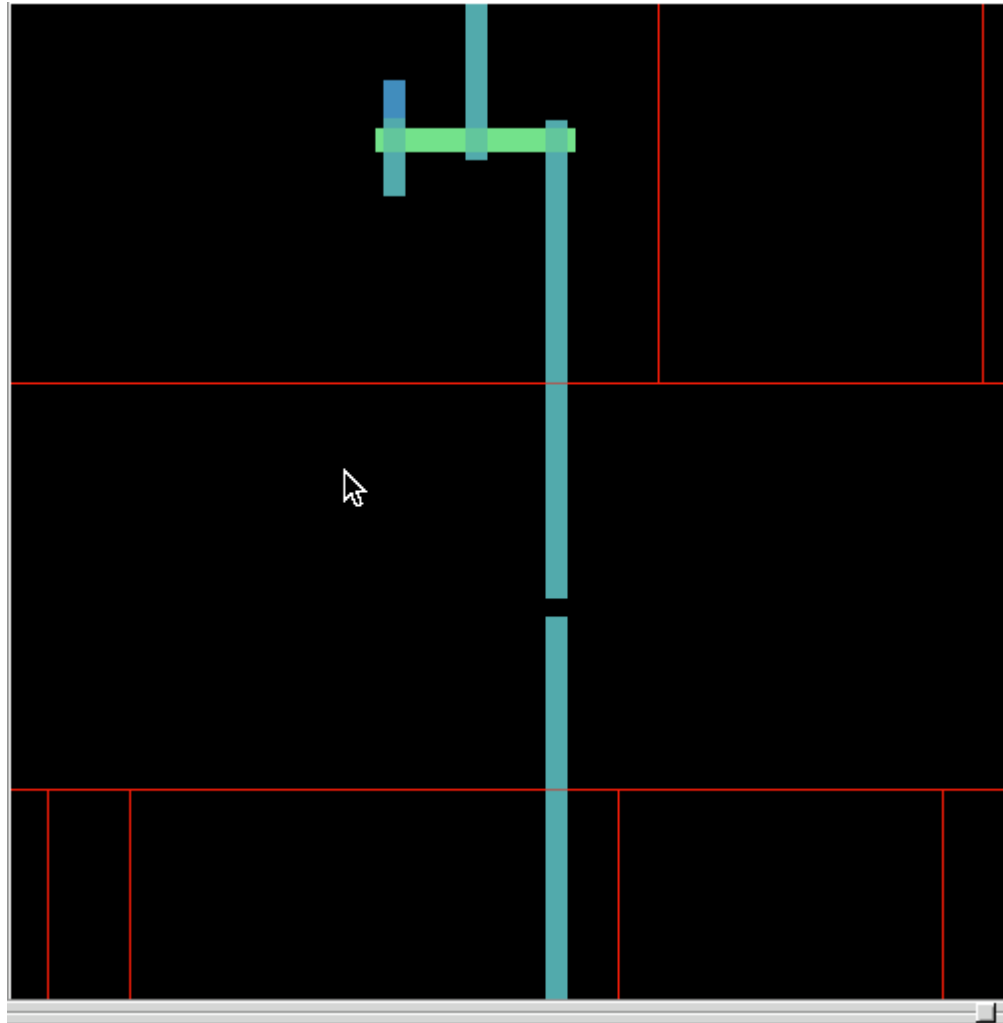
perform analysis –signalEM

...

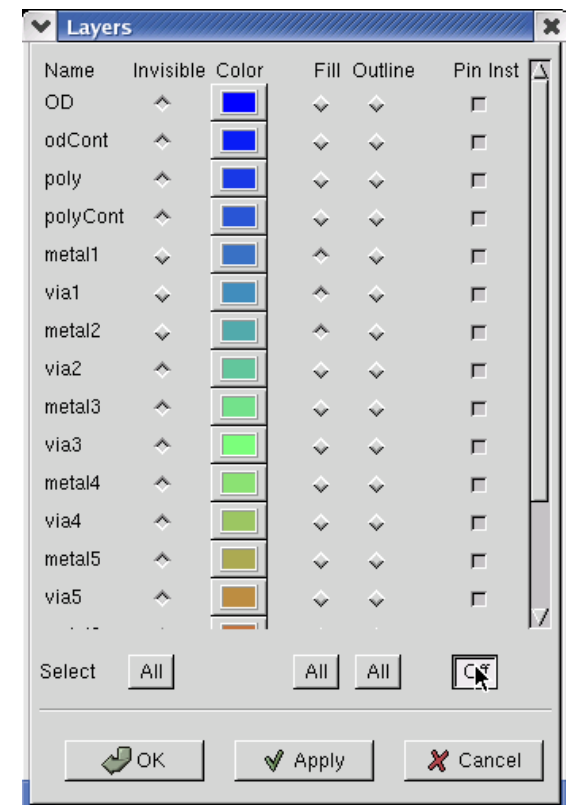
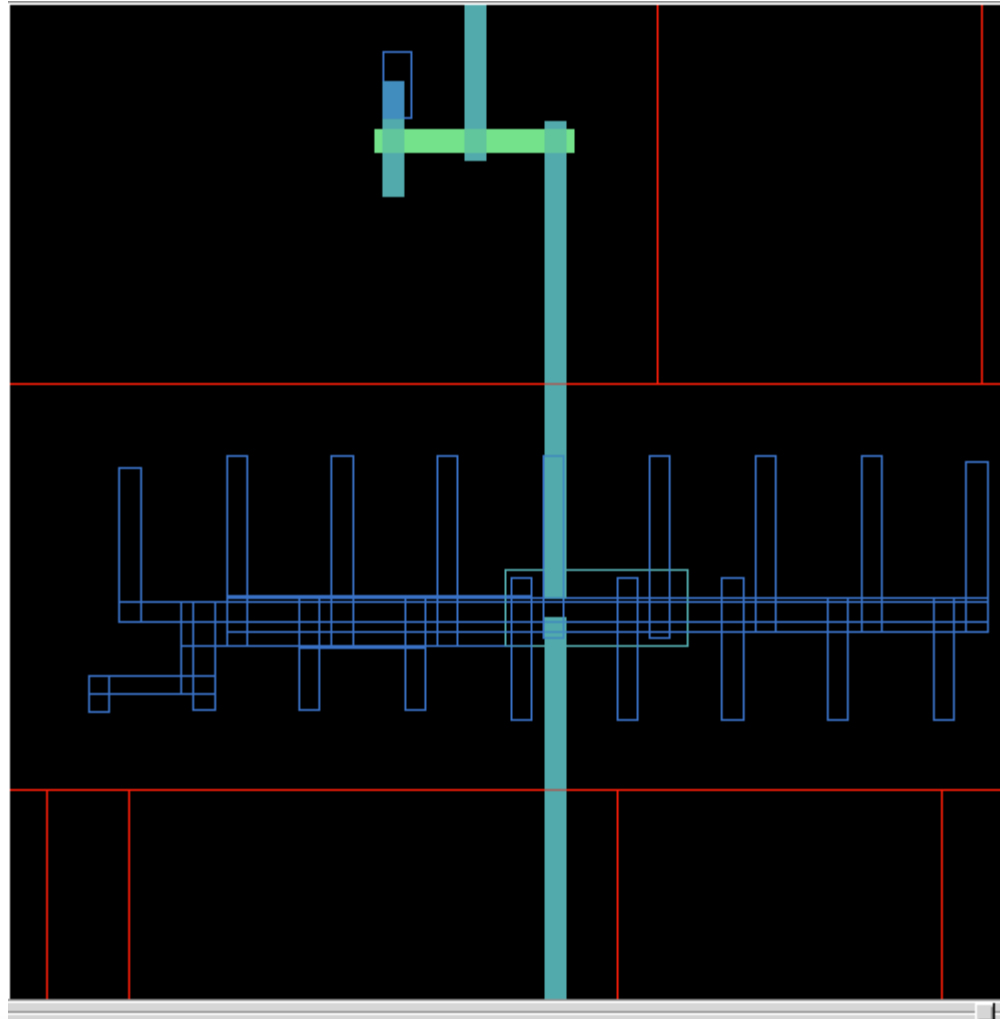
Debugging Tips (*cont'd*)

- **Check Error messages**
 - ANL-400: net not fully-connected
 - NET-184: net not connected to driver
 - WIR-017: extraction problem
 - INT-000
- **Tech file**
 - Missing parameters: dielectric, metal
 - Wrong EM limit
- **GSR keywords to look for:**
 - Additional EM tech files
 - PRIMARY_OUTPUT_LOAD_CAPS
 - DEFAULT_TOGGLE_RATE
 - STA, SPEF etc

Broken wire?



Connected through pin



Data Issue : Incomplete EM Tech Files

- VIA limits only for AVG currents.
- RMS Tech is width based vs avg is a single number.

AVG EM Tech File

```
metal M5 {  
  en 0.896  
}  
  
metal M4 {  
  en 0.896  
}  
  
metal M3 {  
  en 0.896  
}  
  
metal M2 {  
  en 0.896  
}  
  
metal M1 {  
  en 0.896  
}  
  
via RV {  
  en 10.93  
}
```

RMS EM Tech File

```
metal M10 {  
  WIDTH_BASED_EM {  
    width { 0.36 0.66 1.40 2.0 2.50 3.00 4.00 5.00 6.00 7.00 8.00  
    en { 3.835 5.735 9.666 12.602 14.98 17.323 21.949 26.53 31.085 35.625 40.16  
  }  
}  
  
metal M9 {  
  WIDTH_BASED_EM {  
    width { 0.36 0.66 1.40 2.0 2.50 3.00 4.00 5.00 6.00 7.00  
    en { 3.888 5.786 10.039 13.192 15.759 18.297 23.32 28.304 33.268 38.219  
  }  
}
```

Setup Issue:

Default Slews to be provided for nets missing in STA

- For Nets missing STA info, default slew should be defined.

EM_SLEW_NO_STA

Specifies the behavior of signal EM analysis when

- a net is driven by a cell, and
- the net has no STA statistical timing information.

The GSR EM_SLEW_NO_STA keyword has three arguments representing three different ways of modeling STA data:

- 'ideal' (default) - means that the driver is considered ideal, with no internal resistance, so the transition time is very short. So if a net has a significant load (large capacitance) this generates large currents, particularly in PEAK current values.
- 'derived' - means that the transition time is computed the same as if there were no driver driving the net. So any values set by other related GSR keywords-- for example, EM_SLEW_SIG_TIME and EM_SLEW_SIG_PERCENTAGE-- are taken without any changes.
- 'input_transition' - means that the argument from GSR keyword INPUT_TRANSITION (which has a default of 100ps) is used as the value for the missing STA.data. Note the difference in this option compared to 'derived'; the 'derived' value is NOT considered a replacement for STA, but is taking the time argument directly. Fundamentally, the 'input_transition' value still takes the actual net load into account and computes from both the net load and the given INPUT_TRANSITION number its own transition time in a non-linear fashion.

(Note that signal nets that are not driven at all can also be considered, but this keyword is not applicable to them.) *DMP compatible. Optional. Default: ideal.*

Syntax:

```
EM_SLEW_NO_STA [ ideal | derived | input_transition ]
```

Example:

```
EM_SLEW_NO_STA derived
```

Advanced Options in GSR

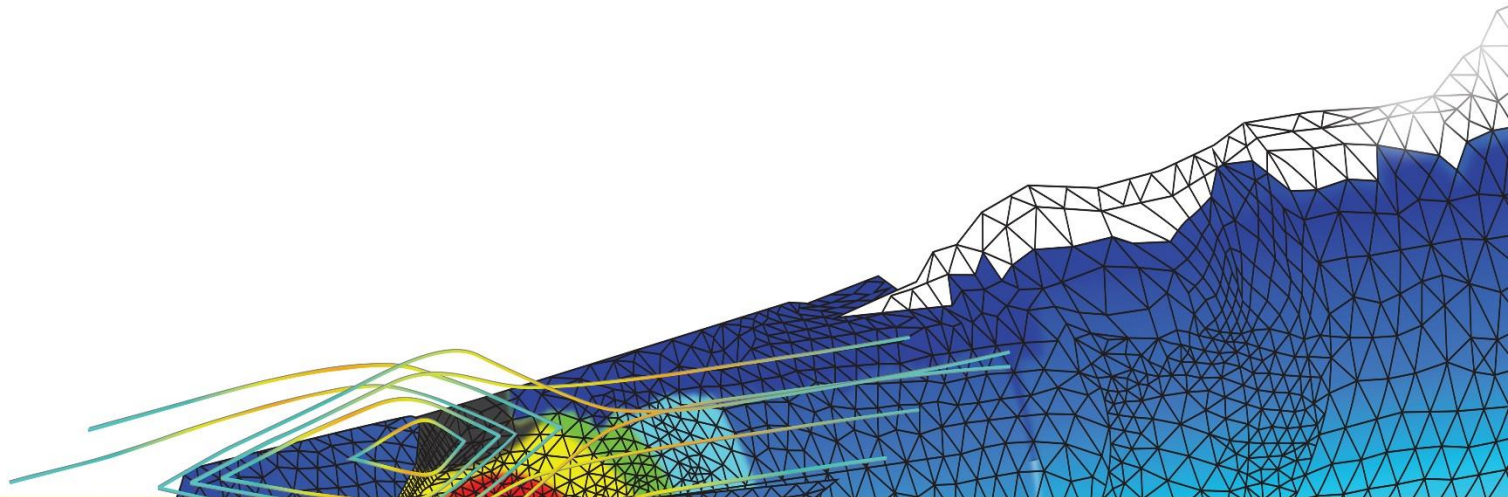
- **EM_WAVEFORM_TYPE**
[Triangle | Polynomial (default)]
- **EM_USE_SLEW [Max | Min | Average]**
- **PRIMARY_OUTPUT_LOAD_CAPS {**
DEFAULT_LOAD 0.1p VDD }
- **EM_REPORT_LINE_NUMBER number**
Default: 1000
- **EM_REPORT_PERCENTAGE number**
Default: 100

Advanced Options in GSR

- **SEM_DEFAULT_PARAMETERS {**
PRIMARY_INPUT_PIN_SLEW <transition_time_ns>
PRIMARY_OUTPUT_PIN_CAP <load_cap_Farads>
CONST_NET_SLEW <transition_time_ns>
CONST_NET_TOGGLE <toggle_rate>
}
- **SEM_IGNORE_NETS_MISSING_DATA**
 - When set to 1, RedHawk ignores nets for which input data (such as slew, frequency, routing and SPEF) is missing or insufficient
- **SEM_RECOVERY_FACTOR <R>**
 - To account for reverse current recovery
- **EM_SLEW_NO_STA < ideal | derived | input_transition >**



RHE – Signal EM



RHE-SEM Overview

- **RHE-SEM enables easier debug of signal EM issues**
- **RHE-SEM includes several capabilities:**
 - Data integrity checks like spef check, Dropped Nets check etc
 - EM hot spot checks
 - Max cap violation checks

Signal EM Summary Page

RedHawk™-Explorer

ANSYS

SummaryData IntegrityDesign WeaknessHotspotsPerformance

BackForward3D-ICFAQ

Design Summary
Results Summary
Power Summary
Static Voltage Drop Summary
Dynamic Voltage Drop Summary
Current w/f Summary
Power EM Summary
Signal EM Summary
PathFinder Summary
Low Power Check Summary
CPM Summary
Run Details

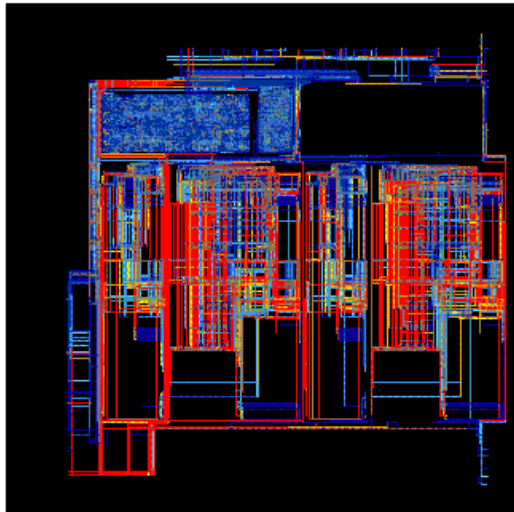
RHE Summary » Signal EM Summary

Signal EM Summary

RMS EMAVG EMPEAK EM

EM Summary

Parameter	Value
Number of Metal EM Violations	8974
Max Metal EM Violation	4953.60%
Number of Via EM Violations	0
Max Via EM Violation	NA
EM Limit Percentage	100 %

Signal EM Map


RHE-SEM Data Integrity Checks

RedHawk™-Explorer

ANSYS

SummaryData IntegrityDesign WeaknessHotspotsPerformance

BackForward3D-ICFAQ

Library Data

LIB Check✓

LEF Check✓

Design Data

SPEF Check✓

STA Check✓

Dropped Nets✗

Dirty Nets✗

RedHawk Data

GSR Tcl Check✗

Log Summary

ERROR Check✗

WARNING Check✗

Data Integrity Summary

Analysis Summary

Input Data	Status	Summary Description
LIB Check	✓	LIB coverage is 98.84 %
LEF Check	✓	LEF coverage is 100.00 %
SPEF Check	✓	SPEF coverage is 95.10 %
STA Check	✓	STA coverage is 99.21 %
Dropped Nets Check	✗	Details
Dirty Nets Check	✗	Details
GSR Tcl Check	✗	Details

Move the mouse pointer above the table rows to highlight the corresponding violation area.

Violation Areas

Following checks are supported:

- LIB check
- LEF check
- SPEF check
- STA check
- Dropped Nets Check
- Dirty Nets Check
- GSR/TCL Check
- Error/Warning Checks

ANSYS

50

RHE-SEM Dropped Nets Check

- **RHE-SEM Dropped Nets check includes**
 - **Missing Driver:** gives a list of Nets with no driver defined in DEF/SPEF
 - **Disconnected Nets:** gives a list of Nets which are not fully connected
 - **Zero Wire Nodes Nets:** gives the list of Nets with Zero wire nodes (nets with tiny/Missing geometries)
 - **Const Nets:** gives the list of CONSTANT nets (Nets which are defined as CONST in STA file. These nets will not be analyzed in SEM)

RHE-SEM Dropped Nets Check: Example

Data Integrity » Dropped Nets

Dropped Nets

Disconnected Nets Check

Missing Driver

Disconnected Nets

Zero Wire Nodes

Const Nets

- Number of nets which are not fully connected or with disconnected receiver: 1484
- Some of the receivers of these net are not connected to the driver or with connectivity issue
- Showing the first 100 nets from adsRHE/adsSEM/reports/disc_receiver.rpt

Disconnected Receiver

Net	Zoom
net_15343	
net_15344	
net_15345	
net_15346	
net_15347	

Related Output Reports »

RHE-SEM Dropped Nets Check:

Related Reports

Data Integrity » Dropped Nets » Related Reports

Dropped Nets

Related Output Reports

Report	Line Count	Definition
adsRHE/adsSEM/reports/missing_driver.rpt	2067	List of nets with missing driver
adsRpt/SignalEM/connectivity_issue.rpt	347	List of nets which are not fully connected
adsRHE/adsSEM/reports/zero_wire_nodes.rpt	0	List of nets with zero wire nodes
adsRHE/adsSEM/reports/const_nets.rpt	0	List of constant nets

Details of Dropped Nets check

RHE-SEM Dirty Nets Check

- **RHE-SEM Dirty Nets check includes**
 - **SPEF Driver** : gives a list of Nets with no driver in SPEF
 - **SPEF Data**: gives the list of Nets which have no data in SPEF
 - **Slew**: gives the list of Nets which have no slew from STA
 - **Pin Cap**: gives the list of nets with missing pin cap

RHE-SEM Dirty Nets Check: Example

Data Integrity » Dirty Nets

Dirty Nets

Missing SPEF Driver Check

SPEF Driver

SPEF Data

Slew

Pin Cap

● Number of nets with no driver in SPEF: 1632

● Showing the first 100 nets from adsRHE/adsSEM/reports/missing_spef_driver.rpt

Missing SPEF Driver

Net	Zoom
net_15572	 
net_15582	 
net_16056	 
net_16359	 
net_16360	 

Related Output Reports »

RHE-SEM Dirty Nets Check:

Related Reports

Data Integrity » Dirty Nets » Related Reports

Dirty Nets

Details of Dirty Nets check

Related Output Reports

Report	Line Count	Definition
adsRHE/adsSEM/reports/missing_spef_driver.rpt	1632	List of nets with missing driver from SPE
adsRHE/adsSEM/reports/missing_spef_cap.rpt	0	List of nets with missing parasitic from S
adsRHE/adsSEM/reports/def_spef_mismatch.rpt	0	List of nets with mismatch between DEF SPEF
adsRHE/adsSEM/reports/missing_receiver.rpt	0	List of nets with missing receiver
adsRHE/adsSEM/reports/partial_disconnection.rpt	0	List of nets with partial disconnection
adsRHE/adsSEM/reports/disconnected_receiver.rpt	0	List of nets with disconnected receiver
adsRHE/adsSEM/reports/missing_frequency.rpt	0	List of nets with missing frequency from
adsRHE/adsSEM/reports/missing_slew.rpt	11	List of nets with missing slew from STA
adsRHE/adsSEM/reports/missing_pin_cap.rpt	3485	List of nets with missing lib pin cap

RHE-SEM GSR/TCL Checks

Data Integrity » GSR TCL Check

GSR TCL Check

● Performs Checks on GSR/TCL Parameters

GSR ::Parameter	Given Value/Total	Recommended Value/Total	
SEM_DEFAULT_PARAMETERS	None 	Const net slew/freq, primary input net slew and primary output load cap can be set using this keyword, which will enable tool to analyze Const nets	
SEM_IGNORE_NETS_MISSING_DATA	0 	This keyword when set to 1, RedHawk drops those net which have missing data for the signal EM analysis	
PRIMARY_OUTPUT_LOAD_CAPS	PRIMARY_OUTPUT_LOAD_CAPS { DEFAULT	This keyword will be used to calculate the Average/RMS current value of the net	
EM_USE_SLEW	Min	Depending on this keyword, RH will use min slew or max slew from STA file	
EM_REPORT_PERCENTAGE	100	EM Percentage threshold used in SEM analysis	
EM_WAVEFORM_TYPE	Polynomial	This keyword is to set the current profile used to estimate the RMS and peak current to triangular or polynomial	
USE_DRAWN_WIDTH_FOR_EM_LOOKUP	0	Specifies that drawn wire width is to be used for the EM rule lookup table, rather than silicon width	
USE_DRAWN_WIDTH_FOR_EM	0	When this option is on, RedHawk uses the drawn wire width to calculate EM current density for wires, rather than silicon width	
EM_SLEW_SIG_PERCENTAGE	0.25	Can be used to specify slew percentage of signal for cells with missing slew and load	
EM_SLEW_SIG_TIME	8e-10	Can be used to specify signal slew for cells with missing slew and load	

Signal EM Hot-spots Section

List of violating Nets

RedHawk™-Explorer

ANSYS

Summary Data Integrity Design Weakness Hotspots Performance

RMS/AVG/PEAK
Checks

Spot Summary » Signal EM RMS Check

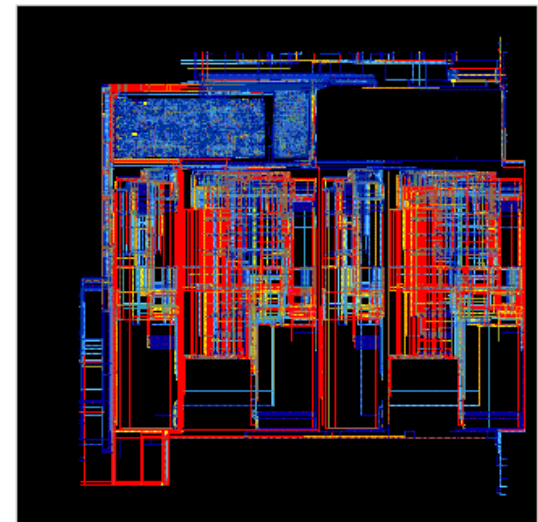
Signal EM RMS Check ?

- Signal EM Hot Spot Exploration
- Highlighting Top 5 Signal EM Hot Spots in the design
- Signal Em Threshold used for PASS/FAIL : 100 %

Top Violations

Net	EM%	Failed nodes	Status
Hot Net : 1	4953.6%	33	✗
Hot Net : 2	2442%	6	✗
Hot Net : 3	2285.05%	11	✗
Hot Net : 4	205.6%	2	✗
Hot Net : 5	1815.1%	32	✗

Worst nets causing
SEM violations
highlighted



List of Worst EM

Show in RedHawk »

ANSYS

Details of the Violating Net : AVG Check

Top Violations

Net	EM%	Failed nodes	Status
Hot Net : 1	6993.11%	29	✗
Hot Net : 2	3448.09%	7	✗
Hot Net : 3	3395.37%	38	✗
Hot Net : 4	2849.88%		
Hot Net : 5	2666.87%		

Hot Spot Summary » Signal EM RMS Check » Hotnet :

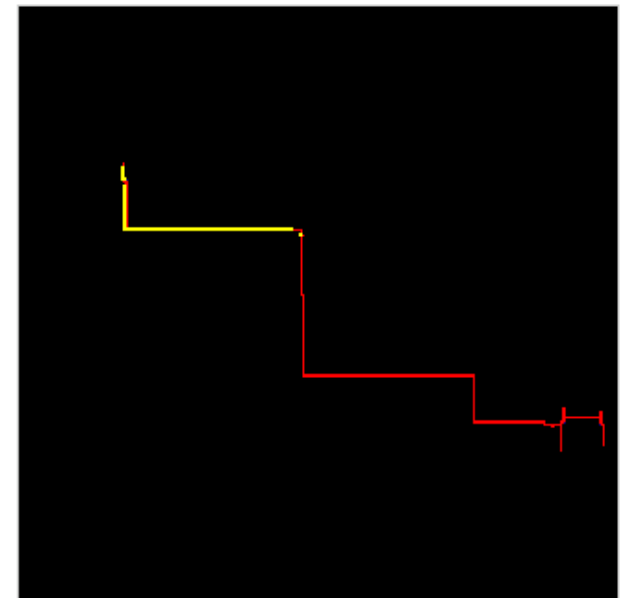
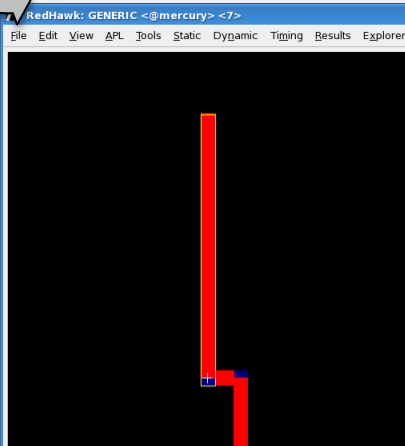
Signal EM Check: RMS-Hot Net

Worst segments
inside the hot-net
highlighted

Top Segments

Segment	EM%	Current(mA)	Layer	Width(u)
Hot Segment: 1	6993.11%	0.024152	METAL2	0.200
Hot Segment: 2	6943.13%	0.0239793	METAL2	0.200
Hot Segment: 3	6823.26%	0.0235653	METAL2	0.200
Hot Segment: 4	6286.38%	0.0217111	METAL3	0.200
Hot Segment: 5	5849.21%	0.0202013	METAL2	0.200

Cross-probing



Show in RedHawk »

More Details for Hot-spots..

Top Violations

Net	EM%	Failed nodes	Status
Hot Net : 1	6993.11%	29	✗
Hot Net : 2	3448.09%	7	✗
Hot Net : 3	3395.37%	38	✗
Hot Net : 4	2849.88%	34	✗
Hot Net : 5	2666.87%	2	✗

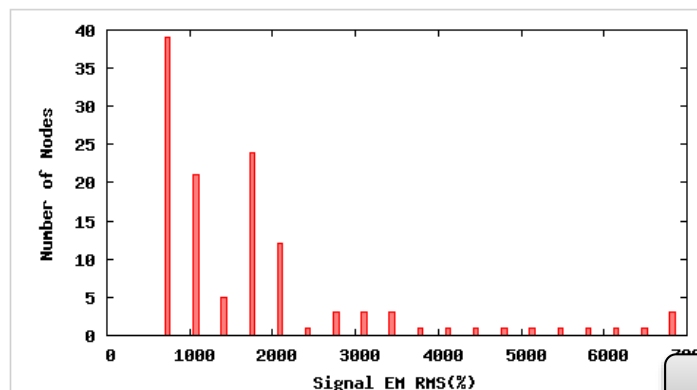
Fanning Net Details

Net	inst_129228/net_74352
Driver Instance	inst_129228/inst_447570
Driver Cell	cell_232
Frequency(Hz)	0
Load(pF)	0.048224
Fanout	7
Driver Slew	3e-10

Net Details

EM Distribution Histogram

Signal EM Distribution Histogram



EM Histogram

Hot Spot Summary » Signal EM RMS Check » Signal EM Check: RMS-Hotnet 1 » Hot Net : 1

Signal EM RMS Check ?

Fanout Details of Net : inst_129228/net_74352

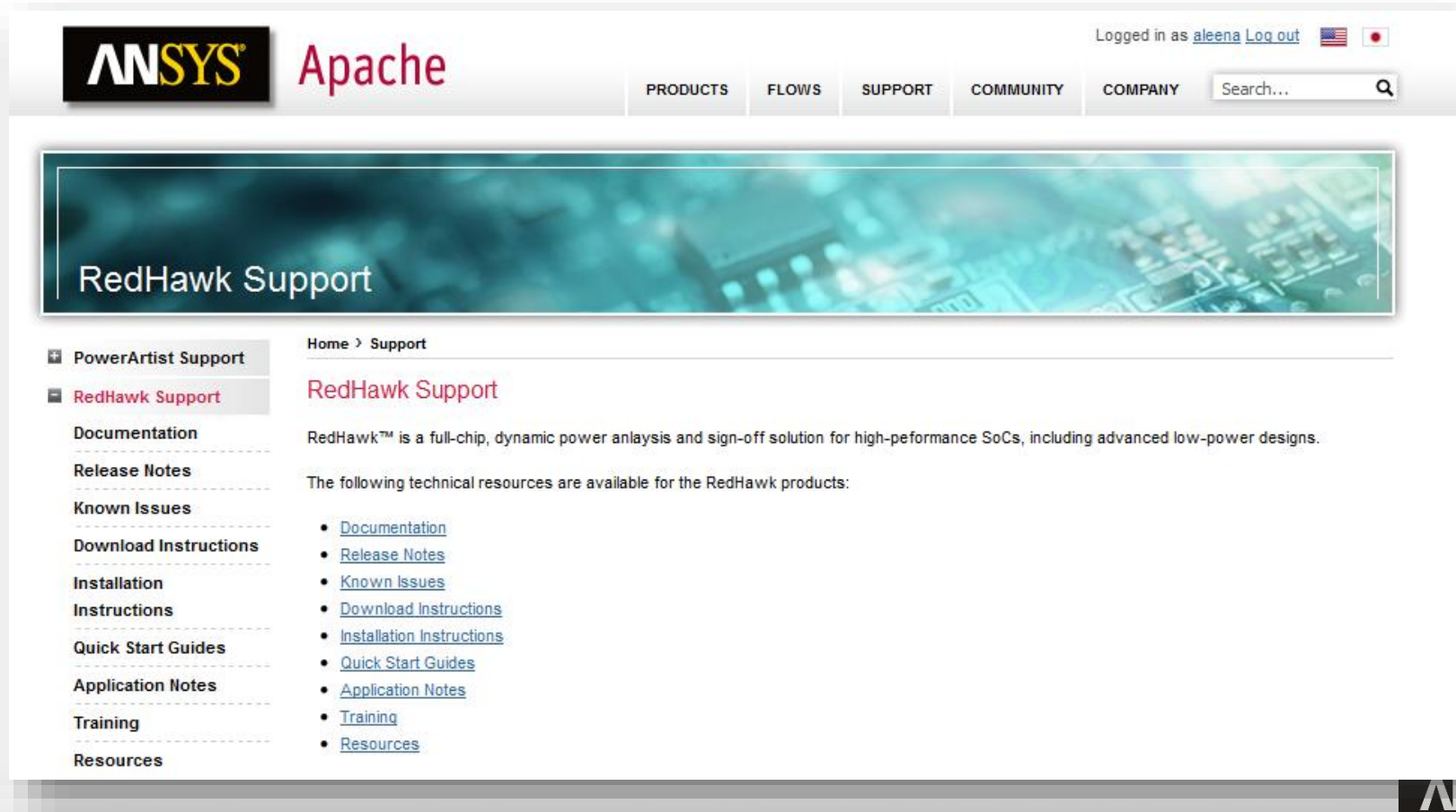
inst_129228/inst_449308
 inst_129228/inst_449307
 inst_129228/inst_449306
 inst_129228/inst_449840
 inst_129228/inst_449838
 inst_129228/inst_449836
 inst_129228/inst_449914

Fanout Details

How to Get Help!

- **Apache Online Customer Support Center**

- <http://support.apache-da.com>
- Email: support@apache-da.com



ANSYS Apache

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RedHawk Support

RedHawk™ is a full-chip, dynamic power analysis and sign-off solution for high-performance SoCs, including advanced low-power designs.

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- [Release Notes](#)
- [Known Issues](#)
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Application Notes

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Resources

Thank You!